

Data Source Authentication Protocol for Aviation Broadband Communication System

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Abstract—The Aviation Broadband Communication System (ABCS) plays a crucial role in the modern civil aviation field. However, when controlling the broadcast information of Broadcast Control Channel (BC), the system faces network security issues such as man in the middle attacks and pseudo base station attacks, which have become increasingly serious. Therefore, when designing and developing aviation broadband communication systems, it is necessary to fully consider the guarantee mechanism for ensuring secure data transmission. This article focuses on the BC control channel and successfully implements source authentication for broadcast control data using the Timed Efficient Stream Loss Tolerant Algorithm Protocol (TESLA) combined with the SM3 algorithm. The proposed method effectively solves the security issues faced by broadcast data. By analyzing the security overhead brought by the added security mechanisms, we have proven that this method is superior to packet-by-packet signature authentication protection and hash chain authentication schemes, thus meeting the requirements of aviation digital communication application scenarios.

Keywords—ABCS, BC, TESLA, source authentication

I. INTRODUCTION

With the rapid increase in civil aviation transportation volume, the volume of aviation communication business is also constantly increasing, posing higher requirements for the performance of modern aviation business communication systems. At present, although ACARS systems and AeroMACS communication systems are widely used, they are difficult to meet the needs of high capacity, high speed, and high security. The future aviation mobile communication technology tends towards large capacity broadband, high performance integrated communication navigation and monitoring systems. The ABCS as a key element, is driving the development of aviation communication from narrowband to broadband. In this context, strengthening the research and development of new generation aviation communication systems, improving communication efficiency and service quality, has become the development trend of aviation mobile communication [1].

In the widespread application of digital communication in civil aviation, the risk of eavesdropping and tampering on unencrypted data in the channel is becoming increasingly prominent. Especially for open aviation data information, there is a potential risk of being maliciously attacked by third parties, leading to unauthorized users stealing data. In order to

ensure the secure of aircraft access, the ground station(GS) needs to continuously broadcast its own parameter information and perform identity authentication. In this scenario, using multicast is more appropriate because using point-to-point symmetric encryption transmission in a unicast environment will consume higher costs and the identity source of data transmission cannot be determined. In order to ensure the security of broadcast information, this article adopts the TESLA protocol to implement source authentication for BC data, in order to prevent man in the middle attacks and information forgery [2].

The rest of the paper is structured as follows. In section II, briefly introduces the research background and related work. The section III introduces the design and functionality of the ABCS model proposed in this paper. The section IV introduces the specific process of the source authentication scheme. The section V conducted testing and security analysis on the specific plan, and compared and analyzed it. The section VI provides a summary of the entire text.

II. RELATED WORK

A. Security issues in aviation data links

ICAO has released the Global Air Navigation Plan (GANP) and the Upgrade of Aviation System Blocks (ASBU), proposing the implementation route for modern air navigation systems [3]. Existing air to ground communication technologies such as ACARS, VHF, CPDLC, and ADS-B are constantly evolving to adapt to the information transmission required for four-dimensional trajectory operation. The protocol stacks of these data links are based on ATN/OSI, gradually evolving into IP based ATN/IPS protocol stacks. However, with the rapid development of civil aviation business, aviation communication business is facing increasingly complex network security issues. Data transmission in open channels is prone to risks such as eavesdropping, forgery, and tampering, and lacks a security authentication mechanism for user entities.

ACARS is used to transmit short messages between AS and GS, and provides a secure communication infrastructure using the AMS [4] framework. CPDLC is used as an air traffic control tool between pilots and controllers, using an authentication protocol based on elliptic curves to achieve identity verification and data integrity protection [5]. ADS-B technology used for air traffic control surveillance faces the challenge of false message forgery. D. McCallie et al. [6] provided a comprehensive overview of the security of ADS-

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B, using asymmetric encryption technology to prevent attackers from abusing ADS-B data and eliminating the need for pre shared keys to be negotiated between aircraft and ground entities. A. Kumar et al. [7] used elliptic curves and RSA algorithm to authenticate WSN data packets packet by packet, QL Huang et al. [8] constructed a sequential hash chain to protect the integrity of data packets on mobile devices. Various authentication methods, such as asymmetric encryption techniques and elliptic curves, are used to protect data integrity, but there are problems such as excessive security overhead or inability to fully resist man in the middle attacks. This article uses the TESLA protocol based on the SM3 algorithm to achieve source authentication of data packets, and achieves the effect of signature through symmetric encryption of delayed public keys. This will effectively solve the security and cost problems generated above.

B. Key Technology

1) SM2 algorithm: It is a public key cryptography algorithm standard [9] based on elliptic curves, and its security reaches the highest level of public key cryptography algorithms. The SM2 algorithm has the same mathematical concept as the ECC algorithm and is an encryption algorithm based on solving the elliptic curve discrete logarithm problem. The elliptic curve is a planar smooth curve composed of the Weierstrass equation, defined as the SM2 elliptic curve on the pseudo Mason prime field $GF(P)$, and its expression is (1).

$$E: y^2 = x^3 + ax + b \quad (1)$$

Among them: $a, b \in GF(p)$, and $4a^3 + 27b^2 \neq 0 \pmod{p}$.

2) SM3 algorithm: It [10] is a hash algorithm that does not require a key. For any message with a text length of no more than 2^{64} bits, a hash value of 256 bits can be generated by padding and iteratively compressing, and the original data cannot be exposed through reverse operations. It is suitable for digital signatures and verifications, message authentication code generation and verification, and random number generation in commercial cryptographic applications. The SM3 algorithm is mainly divided into two steps: filling grouping and iterative compression. Fill in the group, assuming the input length is l bit, first add bit "1" to the end of the message, and then add k "0", where k is the minimum non negative integer that satisfies $l + 1 + k = 448 \pmod{512}$. The filled length is a multiple of 512 and grouped by 512 bits. Iterative compression, iteratively calculating the grouped messages according to (2).

$$V^{(i+1)} = CF(V^{(i)}, B^{(i)}), i = 0, 1, 2, \dots, n-1 \quad (2)$$

Among them: $n = (1 + k + 65) / 52$; CF is a compression function, $V^{(0)}$ is the initial value of 256 bits; $B^{(i)}$ is a message grouping; $V^{(n)}$ is the final hash result.

C. LDACS Protocol Stack

LDACS technology has been continuously evolving [11], and the SESAR project has established corresponding technical standards. It originated from technologies such as B-AMC, TIA902, and WiMAX, and is one of the important wireless networks connecting aircraft and aviation telecommunications networks. It adopts OFDM technology to achieve communication between air and ground through super-frames of forward link (FL) and reverse link (RL), and has similar technical characteristics as 3G and 4G wireless

communication. At present, its net capacity is at least one order of magnitude higher than VDL 2. It can provide secure and reliable data transmission services in terminal areas and high-altitude route areas, and is a communication infrastructure specifically designed for air traffic control (ATC) and air traffic management (ATM) departments. The protocol stack mainly includes physical layer (PHY), medium access layer (MAC), data link service layer (DLS), link management entity (LME), and subnet layer (SNP), as shown in the Fig.1 [12].

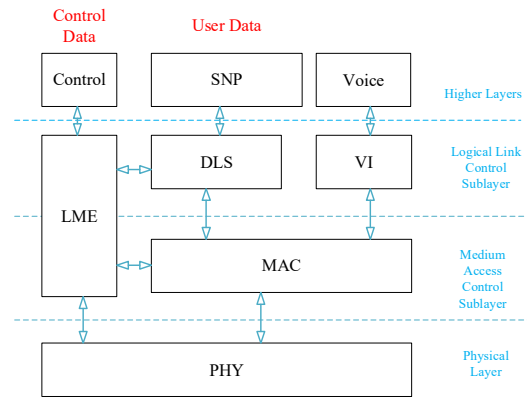


Fig. 1. LDACS protocol stack architecture

It defines three layers of protocol, and the PHY provides a wireless channel transmission method for data between air and ground. The data link layer structure can be divided into two sub layers: the media access control sub layer and the logical link control sub layer. The media access control sub layer only includes the MAC, which is mainly responsible for the mutual mapping between the logical channels of the link layer and the resource blocks of the physical layer; The logical link control sublayer is composed of the DLS and the Voice Interface (VI). The DLS provides service categories for data transmission, promoting multi-purpose concurrent and reliable data transmission; The VI interface provides the function of transmitting and receiving voice data; The LME is mainly responsible for centralized resource management, providing functions such as transmission resource allocation, mobility management, and communication link maintenance; The SNP provides an interface to connect to the aviation telecommunications network, enabling end-to-end data transmission between aircraft, ground stations, and ground control centers.

III. SYSTEM MODEL

A. Architecture Design of ABCS

The main entities of the ABCS include four parts: aircraft station (AS), ground station, ground control center core network (GSC) [13], and security gateway (Sec-GW). The user plane and control plane protocols for communication are integrated between the GS and the core network GSC, ensuring the transmission of data paths between GS and the core network GSC; The communication process between the security gateway Sec-GW and the core network GSC undertakes the functions related to data, entity, and user verification, authorization, billing management, and key management, providing comprehensive protection and protocol management for secure communication between air and ground; The air interface transmission between GS and AS is based on the channels in the forward link (FL) and reverse link (RL) of the physical layer, achieving high

dynamic mobile wireless communication between air and ground. The Air Traffic Service (ATS) and Operations Control Center (AOC) [14] are connected to the GSC, and the

ground network communicates with the aviation network service provider to achieve interaction and coordination between ATC and AOC with the GS, as shown in Fig. 2.

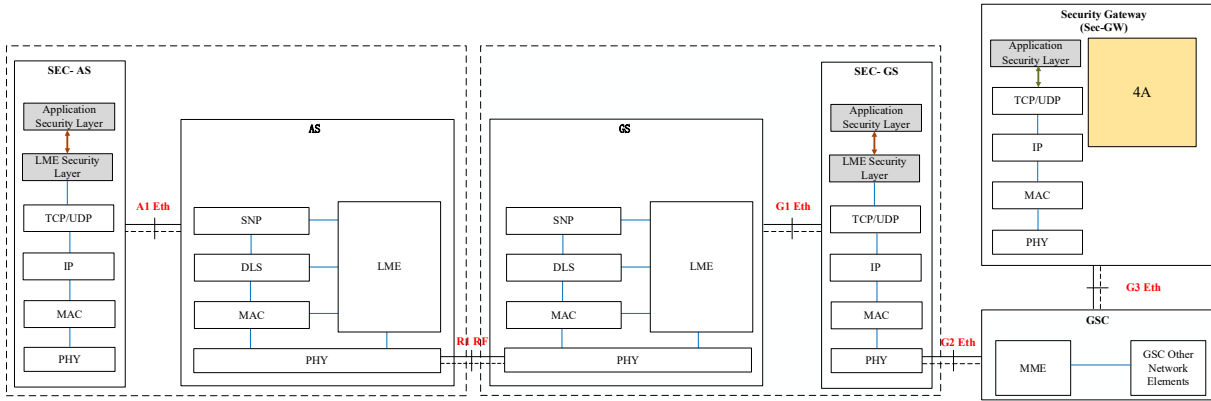


Fig. 2. Protocol stack architecture of aviation broadband communication system

The basic principle of implementing lower level security in the aviation telecommunications network protocol stack is to provide a security layer independent of higher level security functions to strengthen the protection of communication between AS and GS. The design of this security layer aims to provide a deep defense security mechanism for the aviation telecommunications network, as an important component of the ABCS security protection system. Provide the following instructions for the corresponding security module:

1) Application security layer: responsible for the integration of user plane protocols and the transfer of keys, responsible for the encryption, decryption, integrity protection, and verification of user data, and providing third-party authentication processes between entities.

2) LME security layer: integrates control plane protocols, receives security keys transmitted by applications, and ensures the transmission security of control data.

B. Security Objectives

The channels of the ABCS include Broadcast Control Channel (BC), Dedicated Control Channel (DC), Random Access Channel (RA), Common Control Channel (CC), Logical channel (DCH), etc. As shown in Fig.3, the BC channel transmits aircraft broadcast data information that GS continuously moves within the cell, and BC has three subframes; The CC channel transmits data on resource allocation of GS announcing responses to aircraft in the FL link; The DC channel is only used in RL, and the aircraft in the cell send control messages to the GS. Each aircraft has its own specific DC channel, and other users cannot send control information on this channel; The RA channel is used to send access request information for aircraft entering the community.

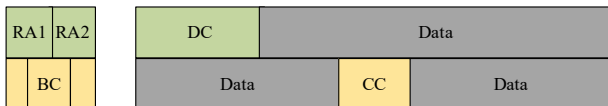


Fig. 3. Physical Layer Frame Structure

Due to the BC channel being used for broadcasting ground parameter messages at GS, AS request access by receiving broadcast messages. However, if an attacker impersonates a GS to broadcast or tampers with broadcast messages, this may result in AS being unable to communicate with the correct GS.

Therefore, source authentication in broadcast messages is crucial to ensure the integrity, authenticity, and availability [15] of BC control messages.

1) Integrity: The data link of the aviation broadband communication system ensures the correctness of the data during transmission, and cannot be modified by unauthorized entities. At the same time, it ensures that information cannot be maliciously tampered with or forged during network communication transmission.

2) Source authentication: Data integrity protection can only ensure that the data has not been forged or tampered with, but the identity of the data initiator cannot be determined. The BC channel of the ABCS should confirm the source identity when broadcasting data.

3) Availability: Ensure that the authorized subject is usable and accessible to the object. If there is a risk of availability, it can lead to network service interruption, suspension, denial of service, and so on.

IV. TESLA SOURCE AUTHENTICATION PROTECTION PROTOCOL BASED ON SM ALGORITHM

A. TESLA protocol

In the previous section, we mentioned a symmetric key mechanism for protecting the integrity of data in a unicast environment, by authenticating data packets through MAC values. However, in the broadcasting environment, the GS of the ABCS uses the BC channel for broadcasting. If the shared key method is still used, there is a risk of attackers impersonating team members to forge data. In addition, symmetric encryption mechanisms incur high overhead. In order to address the issues of packet integrity protection and source authentication in broadcast environments, this paper uses TESLA authentication protocol to protect the BC channel, as shown in Fig.4[16].

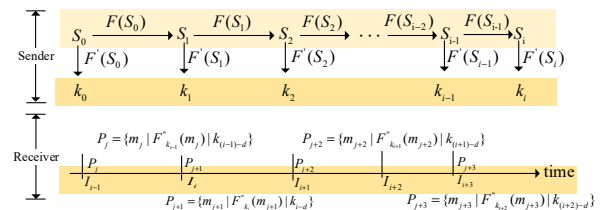


Fig. 4. TESLA Protocol Process

In the authentication protocol of TESLA, a one-way key chain is generated through a pseudo random number S_0 , and the $F(s)$ function generates an authentication key chain $\{S_0, S_1, \dots, S_i\}$ through the hash function SM3. Once again, the encryption key $\{K_0, K_1, \dots, K_i\}$ is derived through the $F'(S_i)$, which is used to encrypt the message cluster $\{m_j, m_{j+1}, m_{j+2}, \dots\}$. At each time interval, the sender can send data packets, and can randomly select a key K_n to calculate

the entire key chain. During this time interval, the data packet $P_j = \{m_j \mid F'_{K_{i-1}}(m_j) \mid K_{(i-1)-d}\}$ is encrypted by a derived key.

B. Scheme Process

In the ABCS network, before entering the target network cell, the GS continuously broadcasts its own beacon messages through the BC channel. These beacon messages contain relevant physical parameters used to establish communication links, thus requiring security protection. For this, we can use the TESLA protocol to encrypt beacon messages and achieve GS data source authentication, as shown in Fig.5.

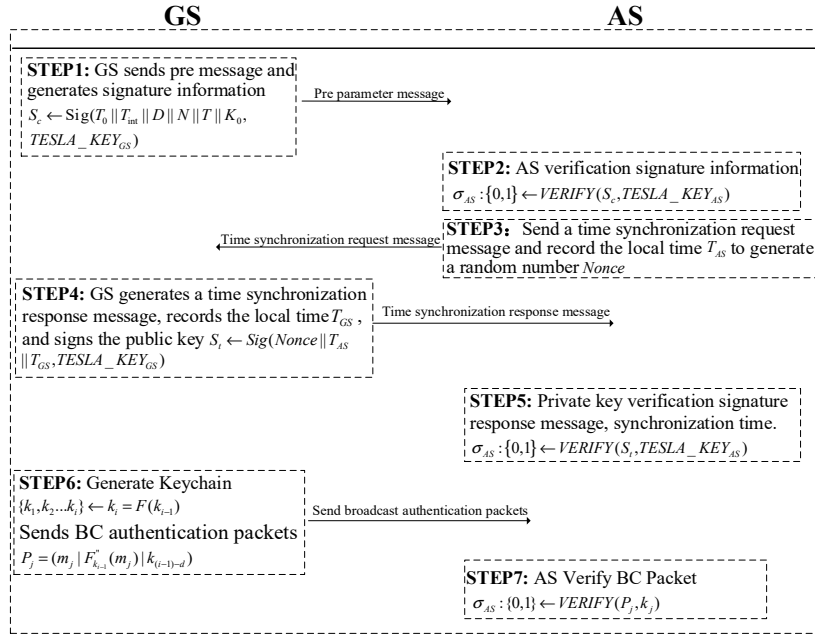


Fig. 5. BC Data Source Authentication Process

1) Protocol initialization. The initialization operation requires multiple specific parameters to be negotiated, including the start time T_0 , interval time T_{int} , key delay disclosure time D , key length N , verification period T , and start key K_0 . The GS broadcasts authentication parameters through the BC channel, but the aircraft at this time does not trust the information sent by the ground. Therefore, the initialization parameters sent by the GS need to be signed with a private key TESLA_KEY_{GS} , and the AS receives the broadcast message and uses the public key TESLA_KEY_{AS} for data verification to ensure the integrity and authenticity of the parameter message.

- GS→AS: GS broadcasts location information at a certain frequency, including its unique identifier UA_{GS} , waiting for the application link connection to enter the broadcasting community;
- AS→GS: AS receives SIB packets broadcasted by GS, queries initialization parameters T_{id} , obtains public and private key pairs $\text{TESLA_KEY}_{GS/AS}$ for signature, sends request packets D_{req} through DC channel, and waits for GS response;
- GS→AS: GS sends a response packet D_{res} through the BC channel, which contains a signed beacon $S_c \leftarrow \text{Sig}(T_0 \parallel T_{\text{int}} \parallel D \parallel N \parallel T \parallel K_0, \text{TESLA_KEY}_{AS})$;

- AS→GS: AS received parameter packet D_{res} for verification $\sigma_{AS} \leftarrow \text{VERIFY}(S_c, \text{TESLA_KEY}_{AS})$.

2) Clock synchronization mechanism. The time synchronization between the GS and the AS require some additional work, so it is necessary to calculate the round-trip transmission time of the data in order to achieve loose synchronization time. This time synchronization does not need to be very accurate, ensuring that it is within an error range Δ .

- AS→GS: AS sends a time synchronization request, records the local time T_{AS} , randomly generates a *Nonce* and sends it to prevent message replay.
- GS→AS: Public key generation time synchronization response packet $S_i \leftarrow \text{Sig}(T_{AS}, T_{GS} \parallel \text{Nonce}, \text{TESLA_KEY}_{GS})$. GS records local time and sends packets using public key TESLA_KEY_{GS} signature.

AS verifies the time signature data packet sent by GS, $\sigma_{AS} : \{0,1\} \leftarrow \text{VERIFY}(S_i, \text{TESLA_KEY}_{AS})$. By recording the current time T , the upper limit value of the time sent at any time can be calculated: $T - T_{AS} - T_{GS}$. Due to the transmission delay of data during transmission, it is not possible to accurately synchronize the time of both parties. Therefore, only a loose synchronization time is needed, and the error of this delay can be expressed as $\Delta = T_{GS} - T_{AS}$. After the clock

synchronization information is completed, GS can start sending packets at a certain rate at each time interval.

3) Constructing a unidirectional key chain for GS. The keychain is generated through the hash algorithm SM3, using a random number S_0 to calculate the initial key, $S_i = F(S_{i-1})$, $0 \leq i \leq N-1$ then calculates N keys.

4) Transmission of GS broadcast data packets. After completing the above steps, the GS begins to send broadcast data packets. At a specific time interval T_j , assuming the time interval D of the announcement delay is 1, the generated one-way key chain is used to encrypt the data packet $\{P_j = m_j | F_{K_{i-1}}(m_j) | K_{(i-1)-d}\}$. The encryption key used is K_{i-2} . The generated data packet P_j with MAC value and security parameters are sent to the AS through the BC channel.

5) Verification of AS Broadcast Data Packets. When AS receives the key K_j from the previous time interval, it can check whether the key is trustworthy through the one-way key generation function $K_j = F^{i-j}(K_i)$. If the subkey is trustworthy, the generated MAC code will continue to verify the integrity, identity authenticity, and non repudiation of the data packet P_j .

V. EXPERIMENTS AND RESULTS

A. Security Analysis

1) Man in the middle attack and message tampering and forgery. Assuming that an attacker has the ability to intercept and tamper with data transmitted on wireless links, as the attacker cannot obtain a single sub key K_{i+1} or pass through a $K_j = F^{i-j}(K_i)$ obtain one-way key chain, if the control message is tampered with without authorization, the receiving end will use the delayed published sub key $K_{(i-1)-d}$ for decryption P_j to verify the data packet and sub key, If the MAC values of the two do not match, discard them.

2) Replay attack. Due to the loose synchronization timer function of the TESLA protocol, intermediaries are unable to perform replay attacks on control data.

B. Performance Analysis

This experiment uses the TESLA protocol to protect the BC control data in the aviation broadband communication system. In this application scenario, GS continuously sends broadcast messages, and multiple AS members receive messages authenticated by the TESLA protocol. The initial key of the TESLA protocol is provided by key derivatives in key management. The generated one-way key chain is iteratively generated, and each key encrypts the broadcast message sent. Table 1 shows the initial key, iteratively generated partial sub keys, and root keys, all of which have a length of 128 bits.

TABLE I. TESLA PROTOCOL KEY INFORMATION

TESLA Keychain	Key Content
Initial key	f3c17b80285d265811a1677c13e31877
Subkey1	d76c10cf3d422b2df44be6cc67da44991
Subkey2	9b91b5331384b09ebcc423a6c345e145

Root Key	1e1a5bd687d677ffcd8a55bd35baad1e
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We tested the time cost of establishing the TESLA protocol at each stage during GS broadcasting. In the test, we set GS to broadcast a message with a byte size of 1KB to AS for a loop test of 10000 sets of data, and record the average time of each stage. As shown in Fig.6, the overall testing time of this scheme is 53.695ms.

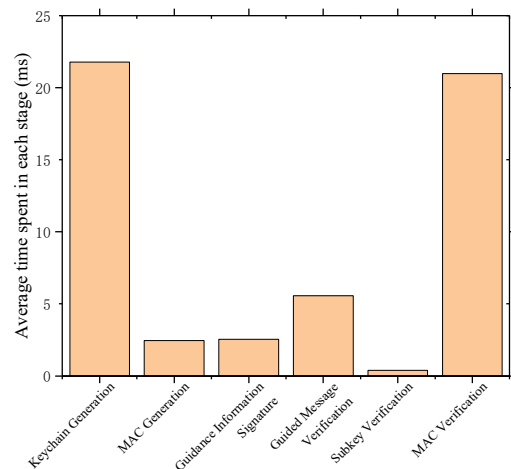


Fig. 6. Average time spent in each phase

Fig.6 shows that the key chain generation and packet MAC verification stages of TESLA account for the largest proportion of time. Due to the preparation of generating a 128 bits one-way keychain in this experiment, a significant amount of time is required during the SM3 operation. Similarly, in the verification process of SM3, due to its special structure, the process of verifying MAC codes takes more time than generating MAC codes.

Meanwhile, we also analyzed the time overhead required for broadcasting control data using TESLA protocol on BC of different sizes. The focus is on analyzing the authentication time required to generate MAC and checksum packets using the TESLA protocol. We have set 8 sets of data, namely 128 Bytes, 256 Bytes, 384 Bytes, 512 Bytes, 640 Bytes, 768 Bytes, 896 Bytes, and 1024 Bytes.

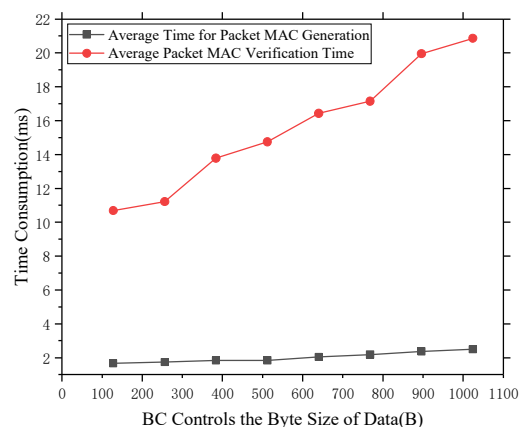


Fig. 7. Time consumption of BC control data

The experimental results are shown in Fig.7, and it can be observed that protecting data on the AS end consumes more time compared to protecting data on the GS end. As the byte size of BC control data increases, apart from the generation time of the key chain, the time cost of protecting data on the GS end will not change significantly and is relatively small.

The reference [7] satisfies the requirements for source authentication and identity authentication of data by using packet by packet signature. The reference [8] uses a message hash chain signature method with an interval of m . This paper compares the time consumption and theoretical values of communication consumption for transmitting n messages between the communication parties with [7] and [8]. Assuming the time consumption T_h for performing a hash function, the total time for performing a signature and verification $T_{sig} + T_{ver}$, the output bit L_h of the hash function, the output bit L_{sig} of the signature, and the L_c guided by TESLA, the signature interval of the message hash chain signature authentication method is m . The performance comparison is shown in Table 2.

TABLE II. PERFORMANCE COMPARISON OF SCHEMES

Authentication Method	Computation Overhead	Communication Overhead
The reference [7]	$2nT_h + n(T_{sig} + T_{ver})$	$n(L_h + L_{sig})$
The reference [8]	$4nT_h + n / m(T_{sig} + T_{ver})$	$2nL_h + n / mL_{sig}$
This paper	$2nT_h + 2(T_{sig} + T_{ver})$	$nL_h + 2L_{sig} + L_c$

TESLA source authentication has a significantly lower number of signatures compared to the reference [7] and [8]. Prior to data source authentication, TESLA needs to negotiate signatures regarding the guiding parameters. The reference [7] each packet needs to be signed one by one, which incurs significant signing overhead. The reference [8] interval packet signatures are used, resulting in significant signing overhead. Although the TESLA protocol performs more hash operations, the computational overhead is significantly reduced compared to signatures. Similarly, in terms of communication overhead, the communication overhead generated by signatures will also be much higher than that of hash operations. Therefore, the use of TESLA protocol for source authentication in aviation broadband communication significantly reduces the overhead.

VI. CONCLUSION

This article proposes a BC control data protection protocol, which combines the TESLA protocol with the domestic hash algorithm SM3 to achieve the authentication effect of symmetric encrypted asymmetric control data. The average total authentication time for each stage of the TESLA protocol applied on the BC channel does not exceed 53.695ms, and it is superior to references 7 and 8 in terms of security overhead, reflecting the effectiveness of the BC channel protection scheme. This protection method provides source authentication for broadcast data, resisting network attacks such as man in the middle and fake base stations, to a certain extent meeting the computational performance of application scenarios, and providing security support for aircraft entering aviation network cell identity authentication and data communication. At present, our work is still based on the application scenario data security protection proposed by theoretical models. In future work, we hope to further optimize the secure transmission protocol from the perspective of reducing security costs and improving security levels, and design security measures suitable for each of them

in complex CC and DC control channels, striving to achieve security protection for all types of data.

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